Where to modify?

1. Set numbers of topics to test

```
[17]: # Set Parameters for LDA model
      number_of_topics = 4
      random_num = 0
```

```
[18]: # Create LDA model for the term frequency document-term matrix
      # Should output general descript about model in the format 'LatentD
      lda_model = LatentDirichletAllocation(n_components=number_of_topics
                                            random_state=random_num)
```

2. TOP_N_TERMS = 15 this is to set up how many tokens that are most related to this topic will show up in our csv file (the spreadsheet)

```
LatentDirichletAllocation
?Documentation for LatentDirichletAllocationiFitted
LatentDirichletAllocation(n_components=4, random_state=0)
```

```
[19]: # topic to top terms
      import numpy as np
      import pandas as pd
      import matplotlib.pyplot as plt
      import os

      TOP_N_TERMS = 15
      OUT_DIR = "outputresults/"
      os.makedirs(OUT_DIR, exist_ok = True)

      terms = np.array(tokenized_data.get_feature_names_out())
```

3. As python to read the csv and treat it as a dataframe: Here, top_n=8 means this will show you 8 tokens that are most closely related to this topic

```
[20]: df = pd.read_csv("outputresults/topic_top_terms.csv")

      # show 8 terms
      top_n = 8
      topics = sorted(df["topic_id"].unique())

      # 4 columns per row
      n_cols = 4
      n_rows = (len(topics) + n_cols -1) // n_cols
```

4. We are then trying to draw the topic prevalence over time, if you have more topics (e.g. you need to modify ```plt.subplot (4,2, figsize=(12,10))``` 4, 2 means four rows and each row has two images)

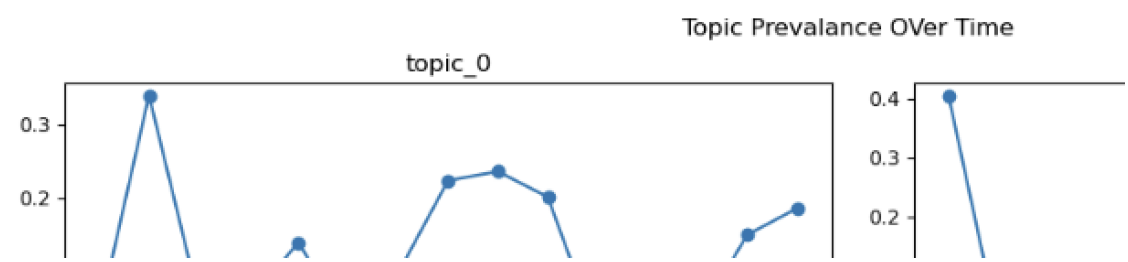
[97]:
```
# see topic prevalence over time

fig, axes = plt.subplots(4, 2, figsize=(12, 10), sharex = True)
axes = axes.flatten()

for ax, topic in zip(axes, topic_cols):
    ax.plot(topic_trends.index.astype(str), topic_trends[topic], marker="o")
    ax.set_title(topic)

plt.suptitle("Topic Prevalance Over Time")
plt.xticks(rotation=45)
plt.tight_layout()
fig.savefig(
    "topic_prevalance.png",
    dpi=300,
    bbox_inches="tight"
)

plt.show()
```



5. MOST IMPORTANT ONE:
   Pick your topic and just switch this one line ```topic=topic_6```
   If you want to check topic 8, do ```topic=topic_8```
   You DO NOT need to switch anything else

[99]:
```
topic = "topic_6"
topic_trends[topic].idxmax()
peak_month = topic_trends[topic].idxmax()
peak_month
```

[99]:
```
Period('2025-10', 'M')
```

6. Last code block:

```Idx = top_topic6_docs.index[1]``` switch this from [1] to [2] or [3] to examine the articles that are most closely related to this topic (the topic you set in the previous step). Remember python starts from 0, so the first article would be idx=topc_topic6_docs.index[0]

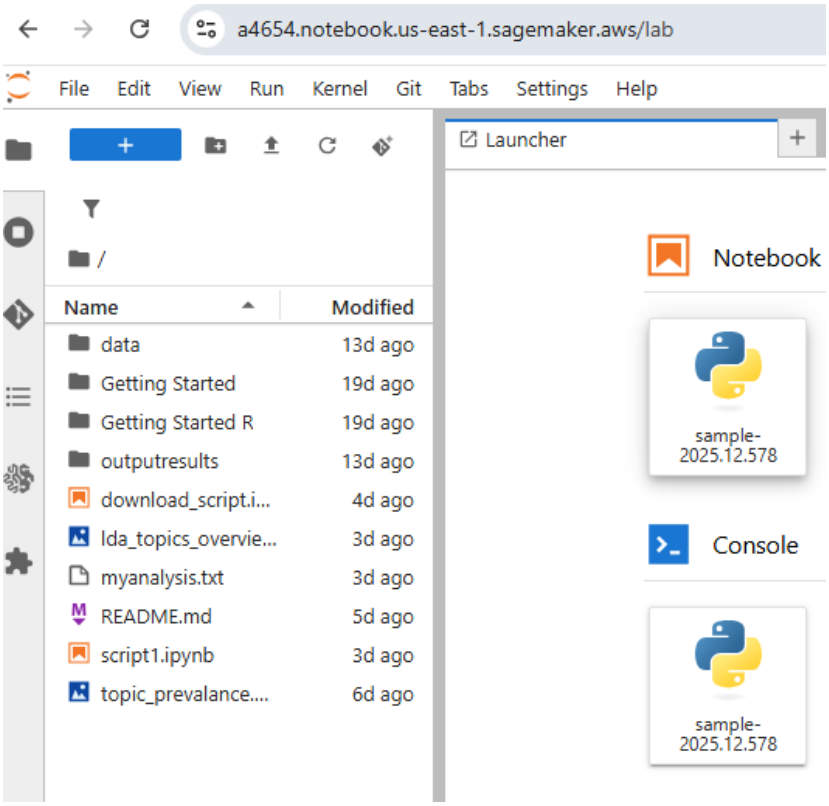
```
idx = top_topic6_docs.index[1]
print(df_metadata_dedup.loc[idx, "date"])
print()
print(df_metadata_dedup.loc[idx, "text"][:5500])
```

```
2025-01-02 00:00:00

Gov. Gretchen Whitmer on Monday signed into law bills that
using discrimination A trio of bills signed by Whitmer wil
tal units to discriminate against current or prospective t
ords from refusing to lease to someone based on how they e
```

Download our data/results:

Start a new script in Jupyter Lab:



Click the + sign on the top left corner→ make sure to select the notebook, “sample-2025.12.578” one on the right hand side.
```

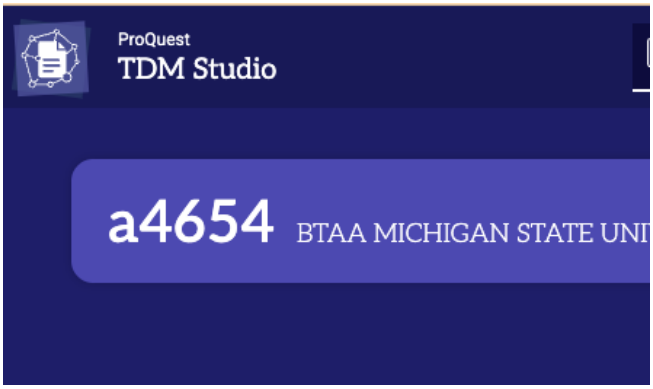
You can name it something like “download_scripts” or really whatever you want, remember the file name.

In the first code block: write this

```
# The file to export
data_to_export = '/home/ec2-user/SageMaker/lda_topics_overview.png'
```

Make sure the file name aligns with how you named your file.
You can write three different blocks with three different files

Write a new code block, copy this code:
NOTE: if your ID is not a4654, swape “a4654” with your own user id:
You can find your user ID either in the ProQuestTDM landing page or in the URL of your Jupyter session:




```
!aws s3 cp $data_to_export s3://pq-tdm-studio-results/tdm-ale-data/a4654/results/
```

```
upload: ../../../lda_topics_overview.png to s3://pq-tdm-studio-results/tdm-ale-data/a4654/results/lda_topics_overview.png
```